TWR Sizing Code

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1 Introduction

For successful completion of the CPLC Lander Challenge, it is essential that we design TOAD with characteristics that provide robust and reliable control and flight. For this reason, developing sizing methods for propellant mass and TWR is essential to our success. This document outlines the process used in the TWR Sizing Code and the results obtained.

2 Sizing Methods

To accomplish this sizing task, we made use of a 3DoF (planar with rotation) simulation of the vehicle with tank dimensions obtained from the tank sizing code developed in parallel with this, and an educated guess for the prop mass needed is used for the tanks to be sized. The sizing code attempts to minimize an arbitrary cost function that combines propellant mass used, flight time, and cumulative deviation from 70% thrust. This latest metric is important as 70% thrust is our point of maximum control authority for the vehicle, giving us important robustness to potential disturbances. To achieve this minimization, we run multiple simulations with TWR values ranging from 1.0 to 2.2. Using the dynamics, an LQR controller is generated and tuned according to the chosen TWR and a reference generator uses PD controllers to create velocity references that creates a constant ascent and descent velocity trajectory. The cost of each trajectory is then recorded and plotted, and the minimum value is identified visually. The cost function weighs all three metrics equally, but this can be tuned to preference and other metrics can be added.

3 Results

This code produced an optimal TWR of 1.44 for the vehicle, using 23.7kg of propellant, and a flight time of approximately 30 seconds. Graphs for the cost function are provided below. Note that the cost of zero indicates a failure, where the minimal thrust at landing produces a TWR greater than one, making landing impossible. The code also produces a Figure showing all the

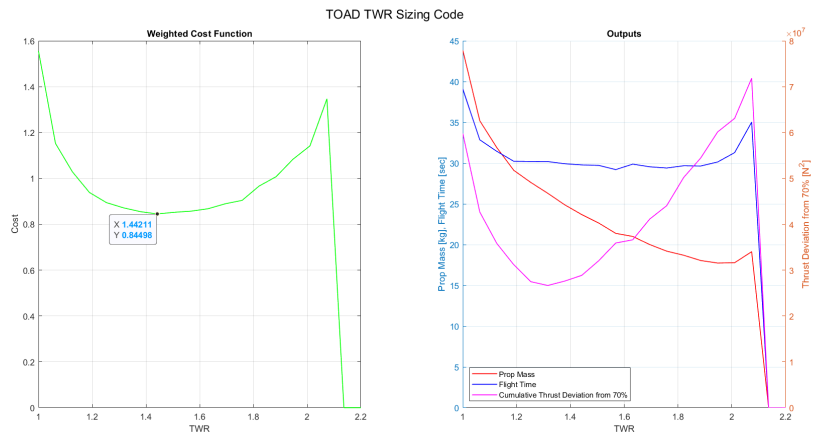

Figure 1: Cost Function Values

generated trajectories, where a lighter color indicates a higher TWR.

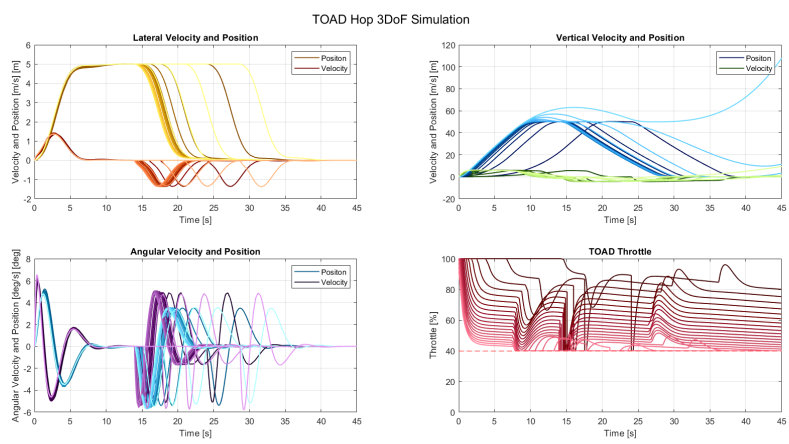

Figure 2: Flight Profiles